

Agricultural Engineer | Dairy Scientist | Applied Livestock Researcher

Profile

Agricultural engineer and dairy scientist with 4+ years of experience in applied livestock research, experimental design, project coordination and statistical analysis. Doctoral research at Leibniz ATB and FU Berlin develops scalable, low-cost gas-sensing systems for livestock emissions monitoring. Experienced in commercial dairy-farm studies, integration of herd-management and environmental data, and communication of scientific results to industry and academic stakeholders.

Professional Experience

Doctoral Researcher (PhD Candidate)

Mar 2023 – present

Leibniz Institute for Agricultural Engineering and Bioeconomy (ATB), Potsdam

Freie Universität Berlin

- Model spatio-temporal dynamics of gaseous emissions in naturally ventilated barns using generalized linear mixed models (GLMM) and time-series analysis in R.
- Architect automated data cleaning, sensor calibration curves, and feature extraction pipelines for continuous high-frequency environmental sensors.
- Integrate heterogeneous herd datasets (milking robots, transponders, feed intake) with greenhouse gas emission metrics to quantify eco-efficiency.
- Engineered and deployed low-cost sensor nodes (NDIR, electrochemical) in harsh barn environments; benchmarked against Gasmet FTIR and Picarro CRDS analyzers.

Scientist (full-time)

Feb 2022 – Feb 2023

Leibniz Institute for Agricultural Engineering and Bioeconomy (ATB), Potsdam

- Maintained and calibrated analytical optical spectrometers (Gasmet FTIR, Picarro CRDS) and continuous sampling manifolds across field campaigns.
- Planned and executed applied research projects on barn climate, air exchange rates, and gaseous emissions under commercial dairy-farm conditions.
- Installed, calibrated, operated, and validated advanced analytical systems including FTIR (Gasmet), CRDS, GC-MS, TDLAS, and PAS.
- Evaluated large-scale multi-sensor experimental datasets, prepared technical reports, and presented findings to industrial and academic partners.

Research Assistant / Master's Thesis

Apr 2021 – Jan 2022

Leibniz Institute for Agricultural Engineering and Bioeconomy (ATB), Potsdam

- Investigated the spatial and temporal distribution of methane, ammonia, and carbon dioxide in a naturally ventilated dairy barn from setup through statistical modeling.
- Evaluated sensor placement strategies to optimize emission quantification and ventilation rate estimations.

Education

PhD in Agricultural Engineering (ongoing)

2023 – present

Freie Universität Berlin

Focus on scalable gas-sensing and spatio-temporal emissions modeling in livestock housing.

M.Sc. Dairy Science (Grade 1.6)

2019 – 2022

Christian-Albrechts-Universität zu Kiel

Ruminant nutrition, herd health, precision livestock farming, forage quality, eco-efficiency, dairy economics.

B.Tech. Dairy Technology

2015 – 2019

College of Dairy Science and Food Technology, India

Dairy processing, quality control, food engineering, dairy microbiology and chemistry.

Key Skills & Expertise

Core Domains: Precision Livestock Farming (PLF), Livestock Housing Technology & Barn Climate, Gaseous Emissions (CH₄, NH₃, CO₂, N₂O), Field Application Science & Technical Support, Environmental Consulting & ESG Reporting, Ruminant Nutrition & Herd Health Analytics

Hardware & Instrumentation: GreenFeed Systems, Gasmet FTIR Analyzers, NDIR & Electrochemical Gas Sensors, CRDS (Picarro), TDLAS & PAS Systems, GC-MS, Climate & Airflow Measurement Instrumentation

Software & Tools: R (tidyverse, lme4, spatstat, ggplot2), Python (pandas, numpy, scikit-learn, beautifulsoup4), MATLAB, LaTeX / XeLaTeX, Git / GitHub, Microsoft Excel / Office

Languages: English (B2 (Professional Working Proficiency)), German (B2 (Professional Working Proficiency)), Hindi (Native)

Leadership, Credentials & Practical Information

- Co-Chair, EurAgEng Young Professionals Network
- PhD Representative, Leibniz ATB
- Communication Manager, Leibniz PhD Network
- 1 First-author peer-reviewed paper, 1 manuscript under review, 8 conference presentations across 9 scientific events

Work Authorisation: Permanent residence in Germany (no sponsorship required)

Driving Licence: Indian Category B, IDP; German Class B practical in progress

References: Prof. Dr. Thomas Amon (Head of Department Sensing and Modelling, ATB Potsdam / Freie Universität Berlin), Dr.-Ing. David Janke (Senior Scientist & Working Group Leader, ATB Potsdam)